INCH-POUND

MIL-B-49430/12C(ER)
1 August 1991
SUPERSEDING
MIL-B-49430/12B(ER)
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MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5093/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

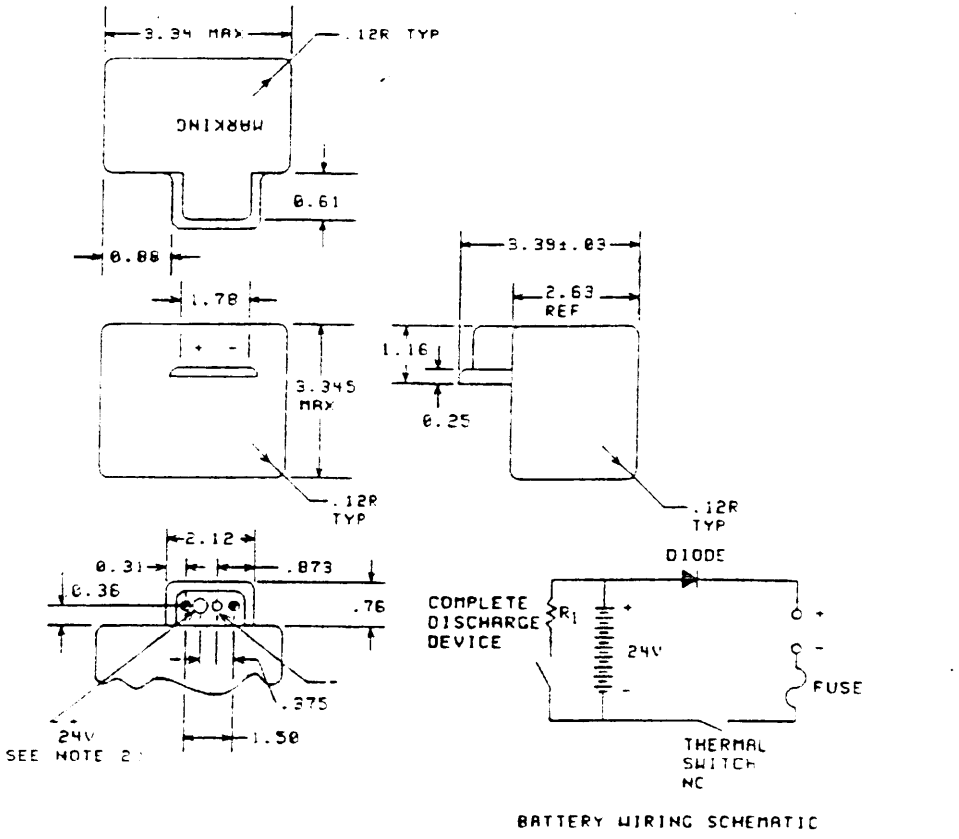


Figure 1. Battery dimensions and wiring schematic.

AMSC N/A

FSC 6135

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NOTES:

1. All dimensions are in inches.

2. Terminal markings to be printed or embossed on jacket in suitable location near the socket.

3. The battery connector shall be a flat, fiberglass socket connector with two (2) push-in terminals per MIL-B-18 as shown in Figure 1.

4. R_1 value shall be between 460 ohms and 650 ohms and have a power rating of 2.0 watts.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 27.45 volts.

Minimum voltage (for capacity): 18.00 volts.

The battery shall contain a 1.25 amperes non-replaceable slow blow fuse in the negative leg.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of the basic specification.

A thermal switch shall be located within the nine cell cluster in accordance with paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

Terminals: Two (2) hole flat plate type (with lead-in countersink on both holes) female socket per MIL-B-18, Type IV, with no obstruction of any holes.

Weight (maximum): 1.43 pounds (650 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 18.00 Volts
I	3.4
L	1.7
H	6.8
HT	5.5
LT	1.6
T	3.1
D	3.1

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Initial voltage delay: During the I, H, HT, T, and D discharge tests, initial voltages below the 18.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests, initial voltage below 18.00 volts cannot exceed a 60.0 second duration.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage: Details on storage conditions for all capacity tests are described in the basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 1.0 ampere constant current at $70\pm5^{\circ}F$ for a minimum of 3.1 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 0.5 ampere constant current for 6.2 hours into voltage reversal.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at 1.0 ampere constant current to zero volts. For the H and HT capacity tests, the batteries shall be discharged through 1.0 ampere constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts. For the I capacity test, the batteries shall be force discharged at 0.5 ampere constant current for 6.8 hours immediately after zero volts is obtained.

Closed-circuit voltage: Load resistance of 48 ohms shall be used between the battery terminals. Voltage shall be 18.0 volts or greater within five seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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